

## DAEN 427 Homework 2 - Alexander Abuabara

Due date: Aug 28, 2025

**Rubric:** any reasonable analysis using Bayesian approach may receive 10 points.

1. Why is it that we identify ourselves at the very beginning of a phone conversation, even to people we already know, but we do not do this when we meet in person? Express your answer within the framework of Bayesian perceptual inference.

**Answer:**

2. When a conversation partner speaks softly, or when a conversation occurs in the presence of significant ambient noise, we sometimes cup our ears and/or look carefully at the speaker's lips. Why, in Bayesian perceptual terms, do we do this?

**Answer:**

3. Design a mini experiment to investigate whether listeners' accuracy, reaction times, and confidence differ when interpreting speech from speakers of different genders and accents. Provide:
- A Bayesian model that frames the research question (in terms of priors, likelihoods, and posteriors).
  - A short paragraph discussing potential biases (that may be implicit and/or represented in the priors).
  - A short paragraph discussing ethical implications of perceptual bias in speech interpretation.
  - A brief methods description, including details about participants, experimental conditions, and materials.
  - Design:
    - A hand-coded sample data table representing how individual participant data might look,
    - A summary results table showing averages across observations.

**Answer:**